Linguagem de Programação para Web I (Javascript)

Métodos de vetores

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Método length

Quantidade de elementos

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>length</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.length); // 3
  </script>
</body>
</html>
```

Método toString()

Transforma em texto

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>toString</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.toString()); // 1,2,3
  </script>
</body>
</html>
```

Método at()

Acessa posição (aceita negativo)

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>at</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.at(0)); // 1
    console.log(arr.at(-1)); // 3
  </script>
</body>
</html>
```

Método join()

Junta com separador

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>join</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.join("-")); // "1-2-3"
  </script>
</body>
</html>
```

Método pop()

Remove último

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>pop</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.pop()); // 3
    console.log(arr); // [1, 2]
  </script>
</body>
</html>
```

Método push()

Adiciona no final

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>push</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.push(4)); // 4
    console.log(arr); // [1, 2, 3, 4]
  </script>
</body>
</html>
```

Método shift()

Remove o primeiro

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>shift</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.shift()); // 1
    console.log(arr); // [2, 3]
  </script>
</body>
</html>
```

Método unshift()

Adiciona no começo

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>unShift</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(arr.unshift(0)); // 4
    console.log(arr); // [0, 1, 2, 3]
  </script>
</body>
</html>
```

Método delete

Remove mas deixa “buraco” ⚠️

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>delete</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3];
    console.log(delete arr[1]); // true
    console.log(arr); // [1, empty, 3]
  </script>
</body>
</html>
```

Método splice()

Remove/adiciona/modifica

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>splice</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.splice(1, 3)); // [2, 3, 4]
    console.log(arr); // [1]
  </script>
</body>
</html>
```

Método concat()

Junta arrays

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>concat</title>
</head>
<body>
  <script>
    let arr1 = [1, 2, 3];
    let arr2 = [4, 5, 6];
    console.log(arr1.concat(arr2)); // [1, 2, 3, 4, 5, 6]
    console.log(arr1); // [1, 2, 3]
  </script>
</body>
</html>
```

Método flat()

Achata arrays

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>flat</title>
</head>
<body>
  <script>
    let arr = [1, [2, 3]];
    console.log(arr.flat()); // [1, 2, 3]
  </script>
</body>
</html>
```

Método slice()

Copia parte

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>slice</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.slice(1, 3)); // [2, 3]
  </script>
</body>
</html>
```

Método map()

Cria novo array transformando

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>map</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.map(x => x * 2)); // [2, 4, 6, 8]
  </script>
</body>
</html>
```

Método forEach()

Só percorre (não retorna)

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>forEach</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    arr.forEach(x => console.log(x)); // 1, 2, 3, 4
  </script>
</body>
</html>
```

Método filter()

Filtra

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>filter</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.filter(x => x > 2)); // [3, 4]
  </script>
</body>
</html>
```

Método reduce()

Acumula valores

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>reduce</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.reduce((soma, x) => soma + x, 0)); // 10
  </script>
</body>
</html>
```

Método every()

Todos passam?

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>every</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.every(x => x > 0)); // true
  </script>
</body>
</html>
```

Método some()

Algum passa?

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>some</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.some(x => x > 3)); // true
  </script>
</body>
</html>
```

Método Spread (...)

Espalha elementos

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Spread</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log([...arr, 5, 6, 7]); // [1, 2, 3, 4, 5, 6, 7]
  </script>
</body>
</html>
```

Método indexOf()

Primeiro índice

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>indexOf</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 4];
    console.log(arr.indexOf(3)); // 2
  </script>
</body>
</html>
```

Método lastIndexOf()

Último índice

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>lastIndexOf</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 3, 4];
    console.log(arr.lastIndexOf(3)); // 3
  </script>
</body>
</html>
```

Método includes()

Existe?

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>includes</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 3, 4];
    console.log(arr.includes(3)); // true
  </script>
</body>
</html>
```

Método find()

Retorna o valor

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>find</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 3, 4];
    console.log(arr.find(x => x > 2)); // 3
  </script>
</body>
</html>
```

Método findIndex()

Retorna índice

```
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>findIndex</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 3, 4];
    console.log(arr.findIndex(x => x > 2)); // 2
  </script>
</body>
</html>
```

Método findLast()

Último valor que bate

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>findLast</title>
</head>
<body>
  <script>
    let arr = [1, 2, 3, 3, 4];
    console.log(arr.findLast(x => x > 2)); // 4
  </script>
</body>
</html>
```

Método findLastIndex()

Último índice

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>findLastIndex</title>
</head>
<body>
  <script>
    let arr = [4, 3, 5, 2, 1];
    console.log(arr.findLastIndex(x => x > 5)); // 2
  </script>
</body>
</html>
```

Método sort()

Ordena

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>sort</title>
</head>
<body>
  <script>
    let arr = [5, 3, 1, 4];
    console.log(arr.sort()); // [1, 3, 4, 5]
  </script>
</body>
</html>
```

Método reverse()

Inverte

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>reverse</title>
</head>
<body>
  <script>
    let arr = [4, 3, 2, 1];
    console.log(arr.reverse()); // [1, 2, 3, 4]
  </script>
</body>
</html>
```

Método toSorted()

Ordena sem alterar original

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>toSorted</title>
</head>
<body>
  <script>
    let arr = [4, 3, 2, 1];
    console.log(arr.toSorted()); // [1, 2, 3, 4]
  </script>
</body>
</html>
```

Método toReversed()

Inverte sem alterar

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>toReversed</title>
</head>
<body>
  <script>
    let arr = [4, 3, 5, 2, 1];
    console.log(arr.toReversed()); // [1, 2, 5, 3, 4]
  </script>
</body>
</html>
```

Método Math.min()

Valor mínimo

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Math.min</title>
</head>
<body>
  <script>
    let arr = [4, 3, 5, 2, 1];
    console.log(Math.min(...arr)); // 1
  </script>
</body>
</html>
```

Método Math.max()

Valor máximo

```html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Math.max</title>
</head>
<body>
  <script>
    let arr = [4, 3, 5, 2, 1];
    console.log(Math.max(...arr)); // 5
  </script>
</body>
</html>
```